

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

6.1.1 ● Random variables X and Y have joint PMF

$$P_{X,Y}(x,y) = \begin{cases} |x+y|/14 & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PMF of $W = X - Y$.

6.1.2 ● For random variables X and Y in Problem 6.1.1, find the PMF of $W = X + 2Y$.

6.1.3 ● N is a binomial ($n = 100, p = 0.4$) random variable. M is a binomial ($n = 50, p = 0.4$) random variable. Given that M and N are independent, what is the PMF of $L = M + N$?

6.1.4 ● Let X and Y be discrete random variables with joint PMF $P_{X,Y}(x,y)$ that is zero except when x and y are integers. Let $W = X + Y$ and show that the PMF of W satisfies

$$P_W(w) = \sum_{x=-\infty}^{\infty} P_{X,Y}(x, w-x).$$

6.1.5 ■ Let X and Y be discrete random variables with joint PMF

$$P_{X,Y}(x,y) = \begin{cases} 0.01 & x = 1, 2, \dots, 10, \\ & y = 1, 2, \dots, 10, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PMF of $W = \min(X, Y)$?

6.1.6 ■ For random variables X and Y in Problem 6.1.5, what is the PMF of $V = \max(X, Y)$?

6.2.1 ● The voltage X across a $1\ \Omega$ resistor is a uniform random variable with parameters 0 and 1. The instantaneous power is $Y = X^2$. Find the CDF $F_Y(y)$ and the PDF $f_Y(y)$ of Y .

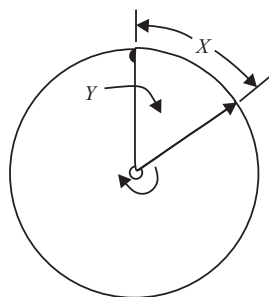
6.2.2 ● X is the Gaussian $(0, 1)$ random variable. Find the CDF of $Y = |X|$ and its expected value $E[Y]$.

6.2.3 ■ In a 50 km Tour de France time trial, a rider's time T , measured in minutes, is the continuous uniform $(60, 75)$ random variable. Let $V = 3000/T$ denote the rider's speed over the course in km/hr. Find the PDF of V .

6.2.4 ● In the presence of a headwind of normalized intensity W , your speed on your bike is $V = g(W) = 20 - 10W^{1/3}$ mi/hr. The wind intensity W is the continuous uniform $(-1, 1)$ random variable. (Note: If W is negative, then the headwind is actually a tailwind.) Find the PDF $f_V(v)$.

6.2.5 ● If X has an exponential (λ) PDF, what is the PDF of $W = X^2$?

6.2.6 ■ Let X denote the position of the pointer after a spin on a wheel of circumference 1. For that same spin, let Y denote the area within the arc defined by the stopping position of the pointer:



(a) What is the relationship between X and Y ?

(b) What is $F_Y(y)$?

(c) What is $f_Y(y)$?

(d) What is $E[Y]$?

6.2.7 ■ U is the uniform $(0, 1)$ random variable and $X = -\ln(1 - U)$.

(a) What is $F_X(x)$?

(b) What is $f_X(x)$?

(c) What is $E[X]$?

6.2.8 ■ X is the uniform $(0, 1)$ random variable. Find a function $g(x)$ such that the PDF of $Y = g(X)$ is

$$f_Y(y) = \begin{cases} 3y^2 & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.2.9 ■ An amplifier circuit has power consumption Y that grows nonlinearly with the input signal voltage X . When the input signal is X volts, the instantaneous power consumed by the amplifier is $Y = 20 + 15X^2$ Watts. The input signal X is the continuous uniform $(-1, 1)$ random variable. Find the PDF $f_Y(y)$.

6.2.10 ■ Use Theorem 6.2 to prove Theorem 6.3.

6.2.11 ■ For the uniform $(0, 1)$ random variable U , find the CDF and PDF of $Y = a + (b - a)U$ with $a < b$. Show that Y is the uniform (a, b) random variable.

6.2.12 ■ Theorem 6.5 required the inverse CDF $F^{-1}(u)$ to exist for $0 < u < 1$. Why was it *not* necessary that $F^{-1}(u)$ exist at either $u = 0$ or $u = 1$?

6.2.13 ♦ X is a continuous random variable. $Y = aX + b$, where $a, b \neq 0$. Prove that

$$f_Y(y) = \frac{f_X((y - b)/a)}{|a|}.$$

Hint: Consider the cases $a < 0$ and $a > 0$ separately.

6.2.14 ♦ Let continuous random variable X have a CDF $F(x)$ such that $F^{-1}(u)$ exists for all u in $[0, 1]$. Show that $U = F(X)$ is the uniform $(0, 1)$ random variable. Hint: U is a random variable such that when $X = x'$, $U = F(x')$. That is, we evaluate the CDF of X at the observed value of X .

6.3.1 ■ X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/3 + 1/3 & -1 \leq x < 0, \\ x/3 + 2/3 & 0 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

$Y = g(X)$ where

$$g(X) = \begin{cases} 0 & X < 0, \\ 100 & X \geq 0. \end{cases}$$

- (a) What is $F_Y(y)$?
- (b) What is $f_Y(y)$?
- (c) What is $E[Y]$?

6.3.2 ■ In a 50 km cycling time trial, a rider's exact time T , measured in minutes, is the continuous uniform $(50, 60)$ random variable. However, a rider's recorded time R in seconds is obtained by rounding *up* T to next whole second. That is, if T is 50 minutes, 27.001 seconds, then $R = 3028$ seconds. On the other hand, if T is exactly 50 minutes 27 seconds, then $R = 3027$. What is the PMF of R ?

6.3.3 ■ The voltage V at the output of a microphone is the continuous uniform $(-1, 1)$ random variable. The microphone voltage is processed by a clipping rectifier with output

$$L = \begin{cases} |V| & |V| \leq 0.5, \\ 0.5 & \text{otherwise.} \end{cases}$$

- (a) What is $P[L = 0.5]$?
- (b) What is $F_L(l)$?
- (c) What is $E[L]$?

6.3.4 ■ U is the uniform random variable with parameters 0 and 2. The random variable W is the output of the clipper:

$$W = g(U) = \begin{cases} U & U \leq 1, \\ 1 & U > 1. \end{cases}$$

Find the CDF $F_W(w)$, the PDF $f_W(w)$, and the expected value $E[W]$.

6.3.5 ■ X is a random variable with CDF by $F_X(x)$. Let $Y = g(X)$ where

$$g(x) = \begin{cases} 10 & x < 0, \\ -10 & x \geq 0. \end{cases}$$

Express $F_Y(y)$ in terms of $F_X(x)$.

6.3.6 ■ Suppose that a cellular phone costs \$30 per month with 300 minutes of use included and that each additional minute of use costs \$0.50. The number of minutes you use the phone in a month is an exponential random variable T with expected value $E[T] = 200$ minutes. The telephone company charges you for exactly how many minutes you use without any rounding of fractional minutes. Let C denote the cost in dollars of one month of service.

- What is $P[C = 30]$?
- What is the PDF of C ?
- What is $E[C]$?

6.3.7 ■ The input voltage to a rectifier is the continuous uniform $(0, 1)$ random variable U . The rectifier output is a random variable W defined by

$$W = g(U) = \begin{cases} 0 & U < 0, \\ U & U \geq 0. \end{cases}$$

Find the CDF $F_W(w)$ and the expected value $E[W]$.

6.3.8 ■ Random variable X has PDF

$$f_X(x) = \begin{cases} x/2 & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

X is processed by a clipping circuit with output

$$Y = \begin{cases} 0.5 & X \leq 1, \\ X & X > 1. \end{cases}$$

- What is $P[Y = 0.5]$?
- Find the CDF $F_Y(y)$.

6.3.9 ■ Given an input voltage V , the output voltage of a half-wave rectifier is given

$$W = \begin{cases} 0 & V \leq 0, \\ V & 0 < V < 10, \\ 10 & V \geq 10. \end{cases}$$

Suppose the input V is the continuous uniform $(-15, 15)$ random variable. Find the PDF of W .

6.3.10 ■ The current X across a resistor is the continuous uniform $(-2, 2)$ random variable. The power dissipated in the resistor is $Y = 9X^2$ Watts.

- Find the CDF and PDF of Y .
- A power measurement circuit is range-limited so that its output is

$$W = \begin{cases} Y & Y < 16, \\ 16 & \text{otherwise.} \end{cases}$$

Find the PDF of W .

6.3.11 ♦ A defective voltmeter measures small voltages as zero. In particular, when the input voltage is V , the measured voltage is

$$W = \begin{cases} 0 & |V| < 0.6, \\ V & \text{otherwise.} \end{cases}$$

If V is the continuous uniform $(-5, 5)$ random variable, what is the PDF of W ?

6.3.12 ♦ X is the continuous uniform $(-3, 3)$ random variable. When X is passed through a limiter, the output is the discrete random variable

$$\hat{X} = g(X) = \begin{cases} -c & X < 0 \\ c & X \geq 0 \end{cases}$$

where c is an unspecified positive constant.

- What is the PMF $P_{\hat{X}}(x)$ of $\hat{X}$?
- When the limiter input is X , the distortion D between the input X and the limiter output $\hat{X}$ is

$$D = d(X) = (X - g(X))^2.$$

In terms of c , find the expected distortion $E[D] = E[d(X)]$. What value of c minimizes $E[D]$?

- (c) Y is a Gaussian random variable with the same expected value and variance as X . What is the PDF of Y ?
- (d) Suppose Y is passed through the limiter yielding the output $\hat{Y} = g(Y)$. The distortion D between the input Y and the limiter output $\hat{Y}$ is

$$D = d(Y) = (Y - g(Y))^2.$$

In terms of c , find the expected distortion $E[D] = E[d(Y)]$. What value of c minimizes $E[D]$?

6.3.13♦♦ In this problem we prove a generalization of Theorem 6.5. Given a random variable X with CDF $F_X(x)$, define

$$\tilde{F}(u) = \min \{x | F_X(x) \geq u\}.$$

This problem proves that for a continuous uniform $(0, 1)$ random variable U , $\hat{X} = \tilde{F}(U)$ has CDF $F_{\hat{X}}(x) = F_X(x)$.

- (a) Show that when $F_X(x)$ is a continuous, strictly increasing function (i.e., X is not mixed, $F_X(x)$ has no jump discontinuities, and $F_X(x)$ has no “flat” intervals (a, b) where $F_X(x) = c$ for $a \leq x \leq b$), then $\tilde{F}(u) = F_X^{-1}(u)$ for $0 < u < 1$.
- (b) Show that if $F_X(x)$ has a jump at $x = x_0$, then $\tilde{F}(u) = x_0$ for all u in the interval

$$F_X(x_0^-) \leq u \leq F_X(x_0^+).$$

- (c) Prove that $\hat{X} = \tilde{F}(U)$ has CDF $F_{\hat{X}}(x) = F_X(x)$.

6.4.1● Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6xy^2 & 0 \leq x, y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $V = \max(X, Y)$. Find the CDF and PDF of V .

6.4.2● For random variables X and Y in Problem 6.4.1, find the CDF and PDF of $W = \min(X, Y)$.

6.4.3■ X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & x \geq 0, y \geq 0, x + y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Are X and Y independent?
- (b) Let $U = \min(X, Y)$. Find the CDF and PDF of U .
- (c) Let $V = \max(X, Y)$. Find the CDF and PDF of V .

6.4.4■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} x + y & 0 \leq x, y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = \max(X, Y)$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$ and $f_W(w)$.

6.4.5■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6y & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = Y - X$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$ and $f_W(w)$.

6.4.6■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = Y/X$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$, $f_W(w)$, and $E[W]$.

6.4.7■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = X/Y$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$, $f_W(w)$, and $E[W]$.

6.4.8 ■ In a simple model of a cellular telephone system, a portable telephone is equally likely to be found anywhere in a circular cell of radius 4 km. (See Problem 5.5.4.) Find the CDF $F_R(r)$ and PDF $f_R(r)$ of R , the distance (in km) between the telephone and the base station at the center of the cell.

6.4.9 ■ X and Y are independent identically distributed Gaussian $(0, 1)$ random variables. Find the CDF of $W = X^2 + Y^2$.

6.4.10 ■ X is the exponential (2) random variable and Z is the Bernoulli $(1/2)$ random variable that is independent of X . Find the PDF of $Y = ZX$.

6.4.11 ♦ X is the Gaussian $(0, 1)$ random variable and Z , independent of X , has PMF

$$P_Z(z) = \begin{cases} 1-p & z = -1, \\ p & z = 1. \end{cases}$$

Find the PDF of $Y = ZX$.

6.4.12 ♦ You are waiting on the platform of the first stop of a Manhattan subway line. You could ride either a local or express train to your destination, which is the last stop on the line. The waiting time X for the next express train is the exponential random variable with $E[X] = 10$ minutes. The waiting time Y for the next local train is the exponential random variable with $E[Y] = 5$ minutes. Although the arrival times X and Y of the trains are random and independent, the trains' travel times are deterministic; the local train travels from first stop to last stop in exactly 15 minutes while the express travels from first to last stop in exactly 5 minutes.

- (a) What is the joint PDF $f_{X,Y}(x, y)$?
- (b) Find $P[L]$ that the local train arrives first at the platform?
- (c) Suppose you board the first train that arrives. Find the PDF of your waiting time $W = \min(X, Y)$.

- (d) The time until the first train (express or local) reaches final stop is $T = \min(X + 5, Y + 15)$. Find $f_T(t)$.
- (e) Suppose the local train does arrive first at your platform. Should you board the local train? Justify your answer. (There may be more than one correct answer.)

6.4.13 ♦ For a constant $a > 0$, random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/a^2 & 0 \leq x, y \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

Find the CDF and PDF of random variable

$$W = \max\left(\frac{X}{Y}, \frac{Y}{X}\right).$$

Hint: Is it possible to observe $W < 1$?

6.4.14 ♦ The joint PDF of X and Y is

$$f_{X,Y}(x, y) = \begin{cases} \lambda^2 e^{-\lambda y} & 0 \leq x < y, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PDF of $W = Y - X$?

6.4.15 ♦♦ Consider random variables X , Y , and W from Problem 6.4.14.

- (a) Are W and X independent?
- (b) Are W and Y independent?

6.4.16 ♦♦ X and Y are independent random variables with CDFs $F_X(x)$ and $F_Y(y)$. Let $U = \min(X, Y)$ and $V = \max(X, Y)$.

- (a) What is $F_{U,V}(u, v)$?
- (b) What is $f_{U,V}(u, v)$?

Hint: To find the joint CDF, let $A = \{U \leq u\}$ and $B = \{V \leq v\}$ and note that $P[AB] = P[B] - P[A^c B]$.

6.5.1 ● Let X and Y be independent discrete random variables such that $P_X(k) = P_Y(k) = 0$ for all non-integer k . Show that the PMF of $W = X + Y$ satisfies

$$P_W(w) = \sum_{k=-\infty}^{\infty} P_X(k) P_Y(w - k).$$

6.5.2 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF of $W = X + Y$.

6.5.3 ■ Find the PDF of $W = X + Y$ when X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.5.4 ■ Find the PDF of $W = X + Y$ when X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1 & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.5.5 ■ Random variables X and Y are independent exponential random variables with expected values $E[X] = 1/\lambda$ and $E[Y] = 1/\mu$. If $\mu \neq \lambda$, what is the PDF of $W = X + Y$? If $\mu = \lambda$, what is $f_W(w)$?

6.5.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PDF of $W = X + Y$?

6.5.7 ■ Continuous random variables X and Y have joint PDF $f_{X,Y}(x,y)$. Show that $W = X - Y$ has PDF

$$f_W(w) = \int_{-\infty}^{\infty} f_{X,Y}(y+w, y) dy.$$

Use a variable substitution to show

$$f_W(w) = \int_{-\infty}^{\infty} f_{X,Y}(x, x-w) dx.$$

6.5.8 ♦ In this problem we show directly that the sum of independent Poisson random variables is Poisson. Let J and K be independent Poisson random variables with

expected values α and β , respectively, and show that $N = J + K$ is a Poisson random variable with expected value $\alpha + \beta$. Hint: Show that

$$P_N(n) = \sum_{m=0}^n P_K(m) P_J(n-m),$$

and then simplify the summation by extracting the sum of a binomial PMF over all possible values.

6.6.1 ■ Use `icdfrv.m` to write a function `w=wrw1(m)` that generates m samples of random variable W from Problem 4.2.4. Note that $F_W^{-1}(u)$ does not exist for $u = 1/4$; however, you must define a function `icdfw(u)` that returns a value for `icdfw(0.25)`. Does it matter what value you return for `u=0.25`?

6.6.2 ■ Write a MATLAB function `u=urw(m)` that generates m samples of random variable U defined in Problem 4.4.7.

6.6.3 ■ For random variable W of Example 6.10, we can generate random samples in two different ways:

1. Generate samples of X and Y and calculate $W = Y/X$.
2. Find the CDF $F_W(w)$ and generate samples using Theorem 6.5.

Write MATLAB functions `w=wrw1(m)` and `w=wrw2(m)` to implement these methods. Does one method run much faster? If so, why? (Use `cputime` to make comparisons.)

6.6.4 ♦ Write a function `y=deltarv(m)` that returns m samples of the random variable X with PDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/4 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Since $F_X^{-1}(u)$ is not defined for $1/2 \leq u < 1$, use the result of Problem 6.3.13.